

Project Design Phase
Proposed Solution

Date	03 October 2026
Team ID	SWTID-2026-5622
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Solution:

S. No	Parameter	Description
1.	Problem Statement (Problem to be solved)	In institutional systems, student and faculty records contain sensitive information that must be accessed only by authorized users . Without proper access restrictions, users may view, create, modify, or delete records beyond their permissions. Role-based access alone may not be sufficient to enforce branch-specific record visibility. Therefore, a secure access control mechanism is required to restrict record access based on user roles and field values while allowing administrators to maintain full access
2.	Idea / Solution description	The proposed solution uses ServiceNow Access Control Lists (ACLs) and server-side scripting to control access to the custom table u_institution_details. Users with the authorized role bb1 can read only records where the Branch field is EEE. Separate roles are configured for record operations: bb2 for Create, bb3 for Write, and bb4 for Delete. An administrator is granted full access through the access-control script. The solution combines role validation, field-based conditions, and scripted authorization to enforce controlled access to institutional records.

3.	Novelty / Uniqueness	The solution combines role-based permissions with field-level record filtering and scripted access validation in ServiceNow. Instead of providing identical access to all users with a general role, it checks both user authorization and the Branch field value . Separate permissions for Read, Create, Write, and Delete provide granular control. The design is configurable and can be extended to additional departments, branches, and access policies.
4.	Social Impact / Customer Satisfaction	The solution improves institutional data security by restricting unauthorized access to student and faculty information. Authorized EEE users can access permitted records, while other users are prevented from viewing restricted information. Separate permissions reduce the possibility of unauthorized record creation, modification, or deletion. Administrators retain full control for system management . This improves data confidentiality, access accountability, and user confidence in the institutional system.
5.	Business Model (Revenue Model)	The solution can be implemented as an internal security configuration for educational institutions and organizations using ServiceNow. Its primary value is improved data protection and controlled access management. For organizations requiring customization, additional services may include ACL configuration, role management, security testing, maintenance, and access-policy enhancements.
6.	Scalability of the Solution	The solution can be extended by adding more user roles, branches, tables, and field-based access conditions . Additional ACL rules can be configured for different departments and organizational requirements. The same approach can be applied to other institutional records and enterprise applications using ServiceNow . Future enhancements may include more detailed access auditing, security monitoring, and expanded authorization policies.